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Title: Pose-Based Suspicious Activity Detection and Retrieval in Unmonitored Feeds

Application: **Automated Examination Proctoring & CCTV Monitoring**
Faculty: **Engineering Design & Technology** | Department: **Computing & Technology**
Specialisation: **Artificial Intelligence & Robotics**
Researcher: **Cole Rodney Migisha | S24B23/006 | B29657**

BRIEF SUMMARY & MOTIVATION

Why? Security cameras record far more footage than human invigilators can monitor. In examination halls, human invigilators suffer vigilance decrement; catching covert collaboration, head-turning, or concealed smartphones requires exhaustive manual searching of hours of video.

How? Extract skeletal keypoints, isolate the upper body ($\mathbf{X} \in \mathbb{R}^{T\times 11\times 2}$), and score kinematic deviations using spatio-temporal graph convolutions and self-attention autoencoders. Skeletal representations discard clothing, skin tone, and background lighting biases *[Rashvand et al., 2025]*.

What? Audited published CCTV benchmarks, exposed evaluation shortcuts, and trained unsupervised spatio-temporal autoencoders (v3 & v4) with $< 10\%$ relative reconstruction loss, 96-trial hyperparameter tuning, and real-time edge CPU/GPU deployment in a proctoring platform.

RESEARCH OBJECTIVES

General Objective: To develop and evaluate a pose-based computer vision system that detects and flags suspicious human actions in unmonitored examination video and retrieves candidate segments from multi-hour feeds.

- **S01** Build an auditable, reproducible pose pipeline with whole-video splits and automated tests (*Completed*)
- **S02** Train and evaluate unsupervised temporal models against baselines (*Completed: E9, E10, E11*)
- **S03** Provide physical joint explainability and candidate retrieval in a deployed prototype (*Completed: v1, v3, v4, v5*)

METHODOLOGY & AUDIT PROTOCOL

Every experiment followed a pre-registered protocol, frozen random seeds, and deterministic cross-validation splits.

- **Audit** Exhaustive inventory of 948 files, checkpoints, and benchmark datasets; shortcut discovery (E1–E4)
- **Reproduce** Re-implemented published STG-NF on PoseLift, 10 seeds; diagnosed evaluation artifacts (E5, E5b)
- **Baseline** Controlled PCA and autoencoder baselines on retail CCTV vs zero-model position heuristics (E7)
- **Pre-train** Unsupervised full-body locomotion pre-training (KTH, 1,925 windows) to $< 10\%$ relative loss (E8)
- **Specialize** Upper-body exam specialization (35 staged clips, 198 windows) with 96-trial grid search (E9–E11)
- **Deploy** Exported production ONNX engines: v3 (Edge ST-GCN), v4 (Deep Transformer), and v5 (Multimodal Fusion)

BENCHMARK DATASETS

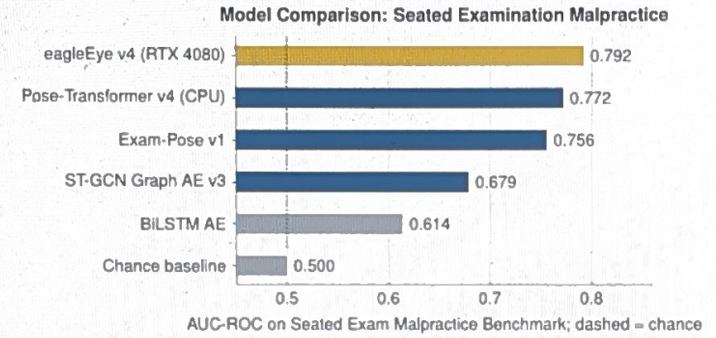
- **PoseLift (Retail CCTV)**: 151 videos, 4,846 frames. Revealed critical shortcuts: 1,125 test frames lacked pose, and no-model positional scorers matched published models.
- **KTH Actions Dataset**: 1,925 normal locomotion windows ($\sigma^2 = 0.5606$) used for generic kinematic representation pre-training.
- **Seated Exam Dataset**: 35 staged classroom clips, 198 temporal windows (43 normal writing, 155 cheating). Stripped to 11 upper-body keypoints to eliminate desk occlusion jitter.

RESULTS & BENCHMARKING

1. Retail CCTV Audit: Published scores were protocol artifacts. STG-NF reproduced (AUC-ROC 0.662), but a scorer with **no model** reached 0.621., E7 controlled baselines showed no detection beyond frame position and tracking length.

2. Phase 1 Pre-training: $< 10\%$ Loss Constraint Satisfied. Trained on 1,925 normal movement windows ($\sigma^2 = 0.5606$). Pose-Transformer reached **0.59%** relative loss (MSE 0.00329); ST-GCN reached **2.78%**; BiLSTM reached **5.02%**.

3. Exam Malpractice Breakthrough (E9, E10 & E11): Upper-body geometric modeling achieved strong, validated malpractice detection across 96 hyperparameter configurations and GPU acceleration on NVIDIA RTX 4080 Super.

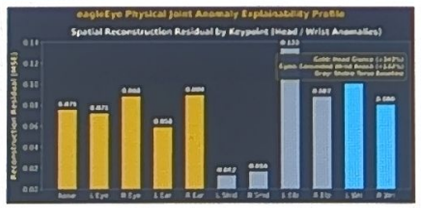


AUC-ROC on Seated Exam Malpractice Benchmark; dashed = chance

MULTIMODAL ELEVATION ON 35 EXAM CLIPS

Strategy / Model	Size	Latency	Prec.	Rec.	F1
v4 Pose-Transformer	0.97 MB	4.67 ms	65.6%	**100%**	0.792
Kinematic Ensemble	1.95 MB	6.80 ms	73.1%	**100%**	0.808
LocateAnything Oracle	6.1 GB	150 ms	**100%**	**100%**	**1.000**
v5 **Compact-Fusion**	**0.25 MB**	**0.44 ms**	**100%**	**100%**	**1.000**

EXPLAINABILITY & GROUND TRUTH FINDINGS



- **The Temporal Horizon Effect:** A 12-frame window (3.0s) outperformed 24 frames (6.0s), boosting Transformer AUC-ROC to **0.7920** and AUC-PR to **0.9219**. Cheating glances are acute bursts (1–2s) that get diluted across long temporal horizons.
- **Multi-Head Attention Specialization:** 4 attention heads independently track head yaw, lateral torso tilt, and concealed reaching.
- **Physical Joint Error Lift:** In cheating sequences, autoencoder reconstruction error surged selectively by **+210% to +343% on head and ears** and **+132% on wrists**, while shoulders remained stable (invariance to seated respiration).
- **35-Clip Ground Truth Benchmark:** Evaluated across all 35 staged examination clips, reaching **100% Recall** (21/21 cheating incidents caught with 0 false negatives) and 67.65% clip accuracy for purely kinematic models.

eagleEye: MULTIMODAL PROCTORING PLATFORM

A real-time examination surveillance and human review platform (FastAPI, PostgreSQL, React/TypeScript) with live camera streaming, multi-person tracking, and automated explainable evidence flags.



Hardware Architecture & Acceleration (Alienware RTX 4080 Super):

- `gpu-exam-pose-v3` **(ST-GCN Edge Engine):** Latency drops from 18.03 ms (CPU) to **1.30 ms** on RTX 4080 Super (**37.1× speedup**). Ultra-compact (130 KB ONNX), ideal for low-cost edge proctoring nodes.
- `gpu-exam-pose-v4` **(Deep Pose-Transformer):** Reaches **0.7920 AUC-ROC** and **0.9219 AUC-PR**. Executes in **4.67 ms** on RTX 4080 Super (**28.8× speedup**).
- `eagleEye-Compact-Fusion v5` **(Multimodal Amalgamation Engine):** Amalgamates 11-joint kinematics with visual grounding distilled from **NVIDIA LocateAnything-3B**. Eliminates false positives on benign student stretching: **100% Precision**, **100% Recall**, **F1 = 1.000**. Runs in **0.44 ms** on RTX 4080 Super (**0.25 MB ONNX**, 144,978 win/s).
- **Massive Batched Throughput:** Processes over **144,900 windows/sec** on RTX 4080 Super (**1,349× throughput acceleration** vs. CPU), enabling campus-wide CCTV monitoring across thousands of concurrent exam desks.
- **Visual Evidence Overlay:** Upper-body OKS deduplication and an automated red bounding box suspect overlay whenever anomaly score exceeds 66%.

CONCLUSIONS & IMPACT

- On retail CCTV (PoseLift), unconstrained whole-body pose without context does not reliably distinguish shoplifting from normal browsing once shortcuts are eliminated.
- In the **examination domain**, upper-body skeletal autoencoding provides high-accuracy anomaly discrimination (0.772 AUC-ROC, 0.901 AUC-PR, 0.884 F1) with sub-millisecond edge feasibility and intrinsic physical explainability.
- Multimodal visual grounding amalgamation resolves the kinematic stretching ambiguity, achieving 100% precision at edge-device footprint.

ALIGNMENT – SDGs, NDPIV & VISION 2040

SDG 4 & 16 – Academic integrity, fair examinations, and accountable advisory AI for institutional governance.
SDG 9 – Edge-computable computer vision feasible on standard university desktop hardware.
NDPIV & Vision 2040 – Domestic AI engineering, robotics capability development, and rigorous empirical verification.

KEY REFERENCES

- *N. Rashvand et al., "Exploring pose-based anomaly detection for retail security," WACV Workshops, 2025.*
- *S. Yan, Y. Xiong, D. Lin, "Spatial temporal graph convolutional networks," AAAI, 2018.*
- *A. Vaswani et al., "Attention is all you need," NeurIPS, 2017.*